

Next, F3 Joins

C1, because:

$$\min[d(F3, F1), d(F3, F4)] = \min(100.14, 120.45) \\ = 100.14$$

So:

$$C1 = \{F1, F3, F4\}$$

Finally, F3 connects to C2 at 100.52, producing one cluster.

Step 2: Min-Max Normalization

The min-max formula is

$$x' = \frac{x - x_{\min}}{x_{\max} - x_{\min}}$$

- Rainfall: $x_{\min} = 280$, $x_{\max} = 520$
- Soil Fertility: $x_{\min} = 3$, $x_{\max} = 9$
- Crop Yield: $x_{\min} = 65$, $x_{\max} = 90$

<u>Farm</u>	<u>Rainfall</u>	<u>Soil Fertility</u>	<u>Crop Yield</u>
		0.8333	0.8000
F1	0.0833	0.1667	0.2000
F2	0.9167	0.5000	0.6000
F3	0.5000	1.0000	1.0000
F4	0.0000	0.0000	0.0000
F5	1.0000		

Step 3: Euclidean Distance

<u>Pair</u>	<u>Distance</u>
	0.2734
F1-F4	0.2734
F2-F5	0.5698
F1-F3	0.6669
F2-F3	0.8124
F3-F4	0.9214
F3-F5	

Step 2: Normalize Order Value

Use Min-max normalization

$$x' = \frac{x - x_{\min}}{x_{\max} - x_{\min}}$$

Here: $x_{\min} = 25,000$

$$x_{\max} = 57,000$$

Therefore:

$$\begin{aligned} O1: \quad & \frac{55000 - 25000}{57000 - 25000} \Rightarrow \frac{30000}{32000} \\ & \Rightarrow \underline{\underline{0.9375}} \end{aligned}$$

$$O2: \quad \frac{25000 - 25000}{57000 - 25000} \Rightarrow \frac{0}{32000} \Rightarrow \underline{\underline{0}}$$

$$O3: \quad \frac{57000 - 25000}{57000 - 25000} \Rightarrow \frac{32000}{32000} \Rightarrow \underline{\underline{1}}$$

$$O4: \quad \frac{30000 - 25000}{57000 - 25000} \Rightarrow \frac{5000}{32000} \Rightarrow \underline{\underline{0.15625}}$$

$$\begin{aligned} O5: \quad & \frac{27000 - 25000}{57000 - 25000} \Rightarrow \frac{2000}{32000} \\ & \Rightarrow \underline{\underline{0.0625}} \end{aligned}$$

Step 1: Calculate distances before normalization (6)

We use Euclidean distance:

$$d(P, Q) = \sqrt{\sum_i (P_i - Q_i)^2}$$

for example,

$$\begin{aligned} d(F1, F4) &= \sqrt{(300-280)^2 + (8-9)^2 + (85-90)^2} \\ &= \sqrt{400 + 1 + 25} \\ &= \sqrt{426} \approx 20.64 \end{aligned}$$

Similarity, the important distances are:

Pair	Distance
F1-F4	20.64
F2-F5	20.64
F1-F3	100.14
F2-F3	100.52
F3-F4	120.45
F3-F5	120.97
F1-F2	200.60
F4-F5	241.31

HAC before normalization

Using Single Linkage the smallest distance is:

$$F1 - F4 = 20.64$$

and

$$F2 - F5 = 20.64$$

Therefore,

$$C_1 = \{F1, F4\}$$

$$C_2 = \{F2, F5\}$$

$$C_3 = \{F3\}$$

Question:

A company clusters employees based on productivity attendance and Skill assessment Scores. Analyze the silhouette Scores obtained for different clustering models and determine the most suitable number of clusters, Justify your answer and explain how the selected model contributes to effective employee performance evaluation.

Given Silhouette Scores

<u>Model</u>	<u>Score</u>
$K=2$	0.48
$K=3$	0.68
$K=4$	0.52

Analysis

The silhouette Score measures the quality of clustering. A higher score indicates better-defined and well-separated clusters.

- $K=2$: Score = 0.48 $\rightarrow$ Moderate clustering
- $K=3$: Score = 0.68 $\rightarrow$ Best clustering
- $K=4$: Score = 0.52 $\rightarrow$ Moderate clustering

Since $K=3$ has the highest silhouette Score (0.68), it is the most suitable model.

Justification:

$K=3$ provides better similarity among employees within the same cluster and better separation between different clusters. It can effectively group employees based on their productivity, attendance and skill assessment.

CSA1603 - Data Warehousing and Mining

CO4 - ATL 1

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Q1 Data Set: Hospital Patient Segmentation

<u>Patient ID</u>	<u>Age</u>	<u>Monthly Medical Expenses (₹)</u>	<u>Health Risk Score</u>
P1	28	12000	65
P2	55	25000	40
P3	42	20000	75
P4	30	11000	70
P5	60	28000	35

SSE values

<u>K</u>	<u>SSE</u>
1	600
2	350
3	220
4	200
5	190

Question:

A healthcare organization uses K-Means clustering to group patients based on their medical expenses and health risk scores. Interpret the following elbow point from the SSE values provided and determine the optimal number of clusters. Discuss how selecting this number of clusters affects patient segmentation and healthcare decision-making.

⇒ For moderate risk :

- 1) Regular follow-up
- 2) Targeted intervention

⇒ For high risk :

- 1) Immediate intervention
- 2) Intensive monitoring (close supervision).

Q2: Dataset : Ecommerce order Analysis

② Dataset A retailer applies hierarchical Agglomerative clustering (HAC) to customer orders using appropriately encoded product category information and standardized order value. A dendrogram is generated. Interpret how clusters are progressively formed and determine the appropriate number of clusters by selecting a suitable cut-off level on the dendrogram. Explain how the selected clusters can help understand customer purchasing behaviour.

Order ID	Product category	order value (₹)
01	Electronics	55000
02	clothing	25000
03	Electronics	57000
04	Furniture	30000
05	clothing	27000

Therefore :

$$\text{No. of clusters} = 2$$

Step 5: Name the clusters

cluster 1 - Highly Engaged , lower-purchase Users

→ U1, U3, U5

cluster 2 - less engaged , high-purchase Users

→ U2, U4

Step-6: Final Result.

$$K = 2$$

$$\text{Cluster 1} = \{U1, U3, U5\}$$

$$\text{Cluster 2} = \{U2, U4\}$$

I analyzed three features - usage, log in, frequency, and purchase amount. The users show two clearly different behavioural patterns. Therefore, the appropriate number of clusters will be formed is "2".

4) Q34 Dataset : Agricultural Crop Monitoring

<u>Farm</u>	<u>Rainfall (mm)</u>	<u>Soil Fertility Score</u>	<u>Crop Yield (tons)</u>
F1	300	8	85
F2	500	4	70
F3	400	6	80
F4	280	9	90
F5	520	3	65

An agricultural research center performs clustering on farm data before and after applying normalization. Interpret the difference observed in the clustering results. Explain how normalization influences cluster formation and improves the reliability of agricultural data analysis.

Order	Electronics	Clothing	Furniture	Normalized Value
01	1	0	0	0.9375
02	0	1	0	0
03	1	0	0	1
04	0	0	1	0.15625
05	0	1	0	0.0625

Step 3: Calculate Euclidean Distances

For two observations:

$$d(P, Q) = \sqrt{\sum_i (P_i - Q_i)^2}$$

01 & 03

$$d(01, 03) = \sqrt{(1-1)^2 + (0-0)^2 + (0-0)^2 + (0.9375-1)^2}$$

$$\Rightarrow \sqrt{0+0+0+(-0.0625)^2}$$

$$\Rightarrow \sqrt{0.00390625}$$

$$d(01, 03) \Rightarrow 0.0625$$

(02 & 05)

$$0.0625$$

$$d(02, 05) = |0 - 0.0625|$$

$$= 0.0625$$

04

$$d(04) = 0.0625$$

The elements within cluster are small so it will be in small cluster only

Question:

A company clusters employees based on productivity attendance and Skill assessment Scores. Analyze the silhouette Scores obtained for different clustering models and determine the most suitable number of clusters, Justify your answer and explain how the selected model contributes to effective employee performance evaluation.

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Since $K=3$ has the highest silhouette Score (0.68), it is the most suitable model.

Justification:

$K=3$ provides better similarity among employees within the same cluster and better separation between different clusters. It can effectively group employees based on their productivity, attendance and skill assessment.

Step 1: Identify the Features

We have 3 features for each user:

- Usage Time (minutes)
- Login Frequency
- Purchase Amount (₹)

Step 2: Observe the Data

Look at the user's behaviour

U1, U3, U5:

- High Usage
- High Login frequency
- Lower purchase amount

So they form one similar group

U2, U4:

- Low Usage
- low login frequency
- High purchase amount

So they form another similar group.

Step 3: Compare the two groups

<u>Features</u>	<u>G1: U1, U3, U5</u>	<u>G2: U2, U4</u>
Usage	High	Low
Logins	High	Low
Purchase	Low	High

There is a clear difference between the two groups.

Step 4: Determine No. of Clusters.

Since there are two distinct behavioral patterns we choose: $K=2$

conclusion:

Best Number of clusters = 3

The Selected $K=3$ model can help the company identify different employee performance groups (such as high, medium, and low performers). This supports better performance evaluation, training, recognition and employee development decisions.

The smallest Distances are F_1-F_4 and F_2-F_5 (7)

Step 4: HAC clustering

At 3 clusters:

$$C_1 = \{F_1, F_4\}$$

$$C_2 = \{F_2, F_5\}$$

$$C_3 = \{F_3\}$$

$\therefore$ Thus similar farms
are grouped
together

Step 5: Interpretation

Before normalization rainfall dominates it
because its numerical range is much larger.

After normalization, all features are scaled to 0-1,
giving rainfall, Soil fertility, and crop yield a more
balanced influence.

~~Now~~

The final 3 clusters:

$$\{F_1, F_4\}, \{F_2, F_5\}, \{F_3\}$$

5) Q5: Dataset : Employee Performance Evaluation

Employee	Productivity	Score Attendance(%)	Skill Assessment Score
E1	88	92	84
E2	62	75	68
E3	78	88	72
E4	55	65	58
E5	93	96	90

Silhouette Scores

Model	Score
$K=2$	0.48
$K=3$	0.68
$K=4$	0.52

Step 4: Using the same matrix notation we are going to find Distance matrix.

id	01	02	03	04	05
01	0	1.697	0.063	1.616	1.663
02	1.697	0	1.732	1.423	0.063
03	0.063	1.732	0	1.647	1.697
04	1.616	1.423	1.647	0	1.1417
05	1.663	0.063	1.697	1.417	0

Step 5: Hierarchical Agglomerative clustering we use Single linkage, where:

$$D(A, B) = \min d(x, y)$$
 between members of the two clusters

Merge 1

Smallest distance:

$$01 - 03 \Rightarrow 0.0625$$

By seeing smallest distance.

$$C_1 = 01 + 03 ; C_1 = \{01, 03\} \Rightarrow 0.0625$$

$$C_2 = 02 + 05 ; C_2 = \{02, 05\} \Rightarrow 0.0625$$

$$C_3 = 04 ; C_3 = \{04\} \Rightarrow 0.0625$$

Merging of cluster can be done by finding the distance between $C_1 \{04\}$, $C_2 \{04\}$

Minimum of $01 \{04\}$, $C_2 04$

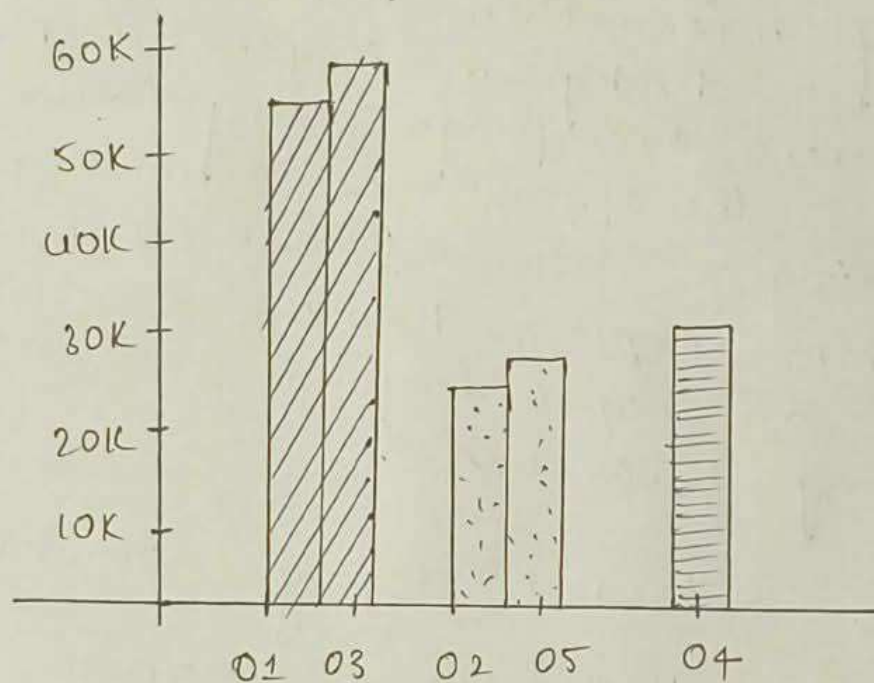
HAC

- It calculates the distance between the clusters
- Merges the two closest clusters
- Recalculates the distance
- Continue until it will become as one cluster.

⇒ 01 & 03 forms 1st cluster - Electronics

⇒ 02 & 05 forms 2nd cluster - Clothing

⇒ 04 forms 3rd cluster - Furniture



But the optimal no. of clusters can not be determine by the given data

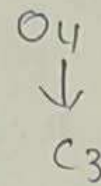
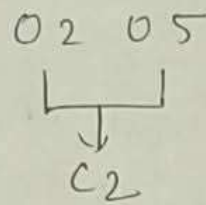
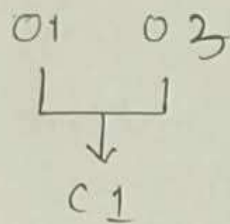
Step:

1) One-hot encoding

order	Electronics	Clothing	Furniture
01	1	0	0
02	0	1	0
03	1	0	0
04	0	0	1
05	0	1	0

$$\begin{aligned}
 & C1 \quad O4 \\
 & (01, 03) \quad O4 \\
 & \min((01, 04), (03, 04)) \\
 & \min(1.616, 1.647) \\
 & \underline{\underline{1.616}}
 \end{aligned}$$

$$\begin{aligned}
 & C2 \quad O4 \\
 & (02, 05) \quad O4 \\
 & \min((02, 04), (05, 04)) \\
 & \min(1.432, 1.417) \\
 & \underline{\underline{1.417}}
 \end{aligned}$$



Q3: Dataset : Mobile App User Behaviour

<u>User</u>	<u>Daily Usage Time (min)</u>	<u>Logic Frequency</u>	<u>In-App Purchase(₹)</u>
U1	120	18	2000
U2	60	5	5000
U3	140	22	1500
U4	50	4	6000
U5	130	20	1800

A mobile application company uses the HAC company to segment users based on their activity patterns. Interpret how users are assigned to clusters using probability-based membership values. Discuss the advantages of soft clustering over hard clustering for customer behaviour analysis.

SSE Means $\rightarrow$ Sum of Squared Errors.

We are using Elbow method

The reduction in SSE in

K	SSE	Reduction from Previous K
1	600	—
2	350	250
3	220	130
4	200	20
5	190	10

We observe that when

$K=2$ & 3 the SSE value is been reduced much and when K value is 4 & 5 the decrease is very small. This variation happens after the value of K is 3 . Therefore Elbow method suggest the value of K is 3

$\therefore$ Based on health risk score 3 clusters have formed.

- i) Low risk - with low expense
- ii) Moderate risk - with moderate expense
- iii) High risk - with high expense.

Impact on Decision making:

$\Rightarrow$ For low risk group:

- i) preventive care
- ii) Routine monitoring